2 **(a)** State Hooke's law.

....

.... [1]

(b) In an experiment to determine the spring constant of a spring X, a student measures the original length of the spring. The student then suspends an object in equilibrium from the spring and measures the final length of the spring.

The spring does not extend beyond its limit of proportionality.

The measurements taken by the student are shown in Table 2.1.

Table 2.1

original length of spring X	(2.1 ± 0.1) cm
final length of spring X	(14.0 ± 0.1) cm
mass of suspended object	(0.300 ± 0.005) kg

Calculate:

(i) the extension, in cm, of spring X together with its absolute uncertainty

extension = (........................ ± ....) cm [1]

(ii) the spring constant of spring X

spring constant = ... $N m^{-1}$ [3]

(iii) the percentage uncertainty in the value of the spring constant calculated in **2(b)(ii)** due to the uncertainties in the measurements.

percentage uncertainty = ... % [2]

(c) An identical object to the one in **2(b)** is suspended in equilibrium from a spring Y that has a greater spring constant than spring X.

Spring Y does not extend beyond its limit of proportionality.

Describe and explain any difference in:

(i) the extension of the springs

....

.... [1]

(ii) the elastic potential energy of the springs.

....

....

....

.... [2]

[Total: 10]